From Land to Liquidity: Land Acquisition Compensation and Elite Mobility in Taiwan's Land Reform

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Outline

Historical Background and Literature Review

Data and Measurement

Short-run baseline population results

Estimating and controlling confounding factors from the land reform

Individual-Level Evidence on Migration

Long-run Outcomes

Conclusion

Appendix

Motivation

- Taiwan's land reform has long been viewed as the root of the "Taiwan Miracle." However, existing studies find little improvement in agricultural output through property rights and only marginal effects through relaxation on restrictions of crop choice.
- In Chinese rural society, landlords often also belonged to the **local elite**, combining land wealth with education, social prestige and political influence (Esherick and Rankin 1990, pp. 310–313).
- Its effects are better understood through the transformation of social structure and the resulting political and economic differentiation.
- Land-to-the-Tiller converted land wealth into liquid assets, facilitating the optimal allocation of people across places and occupations. Different compensation rules for dryland and paddy provide exogenous variation for causal identification.

Outline

Historical Background and Literature Review

Data and Measurement

Short-run baseline population results

Estimating and controlling confounding factors from the land reform

Individual-Level Evidence on Migration

Long-run Outcomes

Conclusion

Appendix

Three stages of postwar land reform

Year	Name	Land or contractual relationship	Administrative action
1949	375 Rent Reduction	Existing Tenant contracts	Rent ceiling, tenure protection, and contract registration
1951	Sale of Public Land	Government-held cultivated land already under tenancy	Sale to incumbent cultivators with installment payment
1953	Land-to-the-Tiller	Privately owned rented land above legal retention limits	Government acquisition and resale to incumbent tenants

Literature: tenancy and agricultural output

Cheung (1969) and Liu (2022)

- Cheung's share-tenancy analysis treats the 37.5% rent rule as a share of realized harvest.
- Liu shows that the ceiling used standard yield. Rent was fixed with respect to realized output.
- The usual sharecropping incentive argument therefore does not describe this rule well.

Kim and Wang (2026)

- They test the direct effect of reform on agricultural output and find little robust gain from the private-land transfer.
- Colonial extension and farmer training had already spread modern techniques. Another yield jump may have been limited.
- Freedom to change crops on former public land may offer a sharper output margin.

Literature: land as a financial and social asset

Koo's recollections

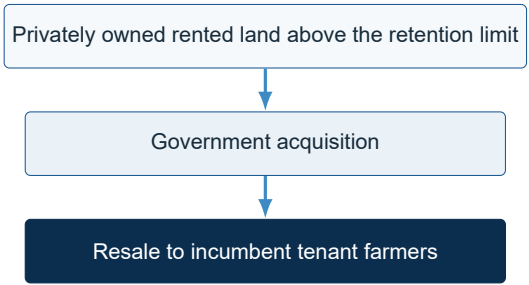
- The account collected in *Jin Han Mei Xiang* emphasizes the release of land wealth into more liquid assets.
- Koo participated closely in reform while standing outside government. His observations provide a distinct participant perspective.

Qu and He

- Qu traces landlord compensation into enterprise shares and industrial finance.
- He reads reform as a change from village-based relations to modern individual property rights and state governance.

This paper studies population reallocation, long-run employment structure, and local electoral outcomes.

The 1953 Land-to-the-Tiller transaction



Compensation to former landlords

- 70% of land value in commodity-denominated land bonds
- 30% in shares of state-owned enterprises
- Bonds paid 4% annual interest and were amortized over ten years
- Outside Penghu, repayment was scheduled twice per year

Statutory text: bond denomination & repayment asset

Original statute: 《制定臺灣省實物土地債券條例》／總統府公報第 383 號／ 1953 年 1 月 26 日

本債券券面實物，分稻穀與甘薯二類：一、稻穀債券，定額十二億六千萬公斤，用以補償徵收田地目部份所包括之兩季田、單季田、看天田、輪作田（包括特殊田）地價。二、甘薯債券，定額四億四千萬公斤，用以補償徵收畑地目部份地價。以上二類債券，由臺灣省政府於民國四十二年度內按照債券面額十足發行。

本債券到期本息，依左列規定償付之：一、稻穀債券，償付稻穀。但用以補償輪作田（包括特殊田）地價所搭發之稻穀債券，按該年期當地稻穀時價折付現金。二、甘藷債券，按該年期之甘藷時價折付現金。前項現金本息，由臺灣土地銀行當地行處經付，其稻穀本息，得轉託糧食局指定之當地倉庫經付，其經付稻穀標準，依臺灣省田賦徵實實收規則之規定。

Both land types: acquisition price = **2.5 times standard annual yield**. 70% in **tax-exempt commodity land bonds**, with **4% annual interest over ten years**.

Dryland	Sweet-potato bonds	Cash repayment at Taiwan Land Bank branches
Paddy land	Paddy bonds	In-kind repayment at designated local warehouses

Bond liquidity and spatial lock-in

Grain procurement and distribution

The Food Bureau organized grain procurement and distribution. Restrictions on private grain transactions limited the liquidity of in-kind paddy repayments.

Emerging securities markets

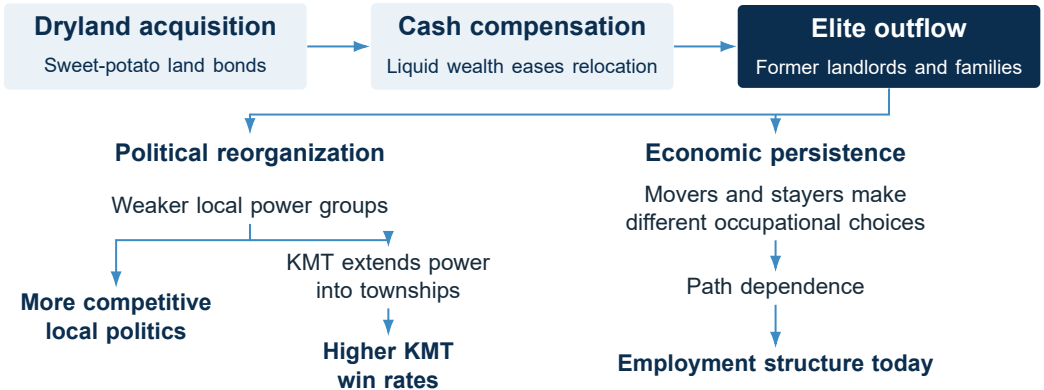
Official and private securities markets began to develop. Cash-paying sweet-potato bonds were easier to discount or pledge in secondary and informal credit markets, at lower transaction costs.

Cash repayment releases the household from spatial lock-in

No local paddy collection or handling for ten years → Immediate relocation of the entire household → Cities or other locations suited to nonfarm work

More liquid compensation facilitates landlords' migration and transition into nonagricultural occupations.

Conceptual framework: cash compensation and elite outflow



Results in brief

- **Population:** -262.6 residents per 10 pp of actual dryland composition in the baseline IV-DID; $-3,489.7$ per 10 pp of actual dryland intensity with both reform exposures controlled.
- **Migration:** $+49.243$ pp in residence outside the birth township per 10 pp of actual redistributed dryland (2SLS).
- **Employment:** -0.979 pp in employers and -1.035 pp in managerial occupations per 10 pp of actual redistributed dryland (2SLS).
- **Elections:** $+0.417$ candidates and -7.089 pp in the winner's vote share per 10 pp of actual redistributed dryland (2SLS).
- **Public land:** -1.090 pp in agricultural-to-nonagricultural transitions per 10 pp of leased public-land share.

Outline

Historical Background and Literature Review

Data and Measurement

Short-run baseline population results

Estimating and controlling confounding factors from the land reform

Individual-Level Evidence on Migration

Long-run Outcomes

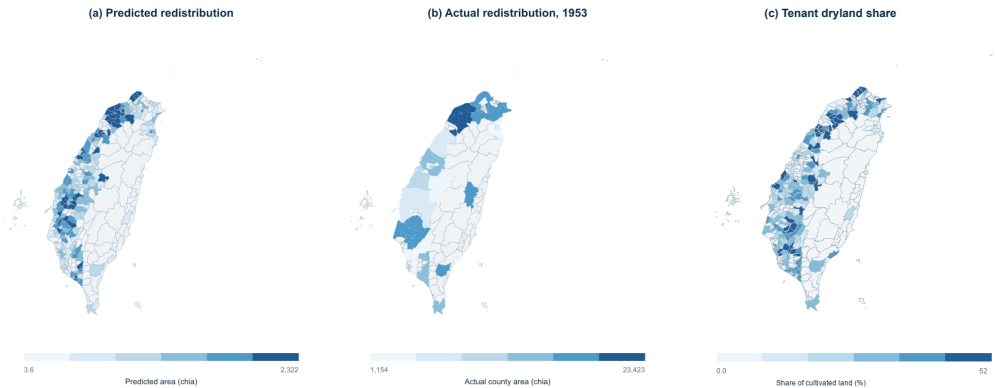
Conclusion

Appendix

Data sources and years

Source	Years
Colonial censuses and population statistics	1915–1935
County and municipal statistical yearbooks, statistical abstracts, household-registration records, and local gazetteers	1947–1954
《臺灣省戶籍統計要覽：民國三十五年至四十七年戶籍統計年報》	1955–1958
《耕地所有權與經營狀況調查報告書》	1950
Digitized research appendix by Chiang, Fan, and Hsu	1950, 1953
County yearbooks: Land-to-the-Tiller actual requisition areas	1953–1957
Luo replication: headmen, settlement diversity, land and tenancy	Colonial period
《農復會工作報告》第五期	1953
Taiwan Social Change Survey	1992–2022 (seven waves)
Manpower Utilization Survey	1978–2000
Internal Migration Survey	1980–1989, 1992
Central Election Commission	1998–2022

Predicted and actual redistribution across space



Research outcomes and land measures

Outcomes

- Registered township population
- Residence outside the birth township
- Employment composition and annual industry transitions
- Township mayoral election outcomes

Land measures and instruments

A_i^D: actual redistributed dryland; A_i: actual redistributed cultivated land.

$$Q_i = \frac{A_i^D}{A_i}, \qquad D_i = \frac{A_i^D}{C_i^{1950}},$$

$$Z_i = \frac{\text{predicted redistributed dryland}}{C_i^{1950}}.$$

$W_i = DryShare_i$: 1950 township dryland share.

Composition models: $Q_i \leftarrow W_i$; intensity models: $D_i \leftarrow Z_i$.

$PublicShare_i$: leased public cultivated land / C_i^{1950}. Actual areas refer to Land-to-the-Tiller requisitions.
Predicted private redistribution is constructed from the owner-size distribution.

Outline

Historical Background and Literature Review

Data and Measurement

Short-run baseline population results

Estimating and controlling confounding factors from the land reform

Individual-Level Evidence on Migration

Long-run Outcomes

Conclusion

Appendix

Endogenous reform intensity and the IV strategy

Actual transfers reflect pre-existing local conditions

- Kim and Wang (2026, Section 4 and Table A.1) find that phase III transfer intensity was positively associated with **pre-reform manufacturing and infrastructure**.
- Pre-reform economic conditions affect both realized redistribution and later outcomes. Regressions on actual transfers can therefore mix reform effects with existing development trajectories.

This paper instruments realized dryland exposure

Actual dryland measure	Pre-reform instrument
Dryland composition, Q_i	1950 township dryland share, W_i
Dryland intensity, D_i	Predicted dryland intensity, Z_i

Z_i combines the 1950 owner-size distribution, statutory retention rules and the tenant dryland mix. In the population panel, each exposure and its instrument are interacted with $Post_t$.

Pre-reform land conditions provide the variation used to instrument realized dryland exposure.

Baseline continuous-treatment difference-in-differences

$$Population_{it} = \alpha_i + \lambda_t + \beta\big(Q_i \times Post_t\big) + \varepsilon_{it}$$

- For Land-to-the-Tiller, $Post_t = 1$ from 1952, when large-scale public communication began.
- Event studies omit 1951, the last year before the first treated year.
- Instrument $Q_i \times Post_t$ with $W_i \times Post_t$, using the 1950 township dryland share W_i. Township and year fixed effects are retained.
- Standard errors are clustered by the 235 treatment units.

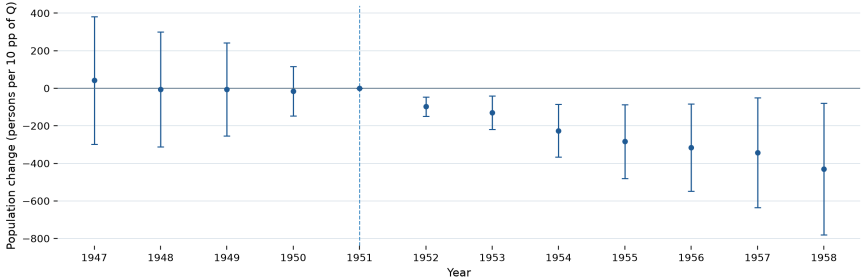
Year	1947	1948	1949	1950	1951	1952	1953	1954	1955	1956	1957	1958
Units	147	148	152	203	235	235	225	235	235	235	235	235

Baseline estimates and sample completeness

	All available (1)	95% coverage (2)	85% coverage (3)	Complete years (4)
$Q_i \times \mathbf{1}[t \geq 1952]$	-262.581^{**}	-262.156^{***}	-255.776^{***}	-282.684^{***}
	(128.740)	(97.632)	(97.769)	(107.476)
p-value	0.041	0.007	0.009	0.009
Township FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Observations	2,520	1,870	2,073	1,645
Treatment units	235	235	235	235
Sample years	1947–1958	1951–1958	1950–1958	7 complete years

Dependent variable: population count. 2SLS: $Q_i \times Post_t$ instrumented by $W_i \times Post_t$. Coefficients per 10 pp of Q_i. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Event-study estimates and pre-trend assessment



2SLS: yearly Q_i interactions instrumented by yearly W_i interactions; 1951 omitted. Error bars: township-clustered 95% intervals. Pre-treatment joint test, 1947–1950: $\chi^2(4) = 4.653$, $p = 0.325$. Post-treatment joint test, 1952–1958: $p < 0.001$.

Outline

Historical Background and Literature Review

Data and Measurement

Short-run baseline population results

Estimating and controlling confounding factors from the land reform

Individual-Level Evidence on Migration

Long-run Outcomes

Conclusion

Appendix

Construction of predicted redistribution

1. Fit the grouped owner-size distribution

$$(\widehat{\mu}_i, \widehat{\sigma}_i) = \arg\max_{\mu,\,\sigma>0} \sum_{b=1}^{12} \left(n_{ib} + 25\pi_{c(i)b}\right) \ln[F_i(u_b) - F_i(l_b)]\,.$$

$$F_i(a) = \Phi\left(\frac{\ln a - \mu_i}{\sigma_i}\right).$$

2. Predict land above the retention threshold

$$\widehat{R}_i = N_i\, \mathbb{E}_{\widehat{F}_i}[(A_i - h)_+] = N_i \int_h^{\infty} [1 - \widehat{F}_i(a)]\, da, \qquad h = 3 \text{ chia}.$$

3. Allocate the dryland component using the tenant land mix

$$R_i = \frac{\widehat{R}_i}{C_i}, \qquad Z_i = R_i \times \frac{Dry_i^{\text{tenant}}}{C_i^{\text{tenant}}}.$$

n_{ib}: owners in size bin b, bounded by l_b, u_b. $\pi_{c(i)b}$: county bin share. N_i: township owner count.

Land-size bins are in hectares. The retention threshold is 3 chia = 2.909751 hectares; predicted area and township cultivated area are converted to the same unit.

The 1951 sale of public land



Institutional distinction

- The 1951 program covered government-held cultivated land.
- Incumbent cultivators purchased the land in installments.
- Its timing predates the 1953 acquisition of privately owned rented land.

Empirical proxy

Leased public cultivated land in the 1950 survey, measured in chia and transformed using a standardized inverse hyperbolic sine.

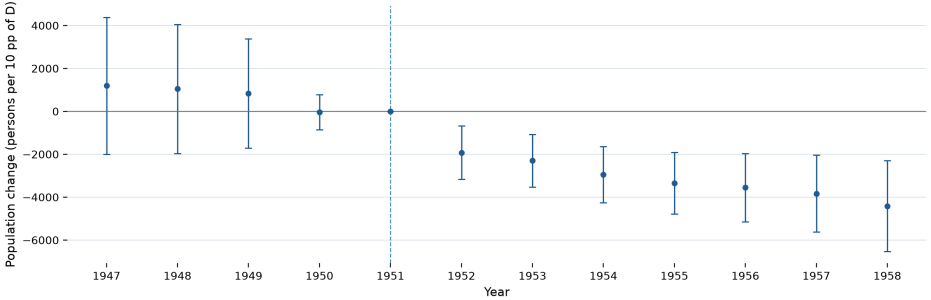
Population effects: actual redistributed dryland

$$Population_{it} = \alpha_i + \lambda_t + \beta(D_i \times Post_t) + \gamma(PublicLease_i \times Post_t) + \delta(R_i \times Post_t) + \varepsilon_{it}.$$

	2SLS dryland intensity	+ Public land	+ Both
$D_i \times Post_t$ (per 10 pp)	-1960.985^{**} (854.909)	-2035.992^{**} (881.383)	-3489.727^{***} (832.503)
$PublicLease_i \times Post_t$ (1 SD)	–	516.777^{***} (170.789)	538.739^{***} (166.741)
$R_i \times Post_t$ (per 10 pp)	–	–	398.398^{***} (151.411)
Township and year FE	Yes	Yes	Yes
Observations	2,520	2,520	2,520
Townships	235	235	235

2SLS: $D_i \times Post_t$ instrumented by $Z_i \times Post_t$. $Post_t = 1$ from 1952. Township-clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Event study with both reform exposures controlled



Pre-treatment joint test $\chi^2(4) = 0.884$, $p = 0.927$

Post-treatment joint test $\chi^2(7) = 25.269$, $p < 0.001$

2SLS: yearly D_i interactions instrumented by yearly Z_i interactions; 1951 omitted. Township-clustered 95% confidence intervals.

Outline

Historical Background and Literature Review

Data and Measurement

Short-run baseline population results

Estimating and controlling confounding factors from the land reform

Individual-Level Evidence on Migration

Long-run Outcomes

Conclusion

Appendix

Migration and migrants' characteristics

Taiwan Social Change Survey, 2012, 2017, and 2022

Actual redistributed dryland and migration (2SLS)

Sample: born before 1953 Outcome	D_i (per 10 pp)	Observations	Years
Residence outside the birth township (pp)	49.243^{*} (26.711)	634	2012–2022
Workplace outside the birth township (pp)	97.226^{**} (32.157)	176	2012–2022

D_i instrumented by Z_i; ten historical controls; county, survey-year, and birth-year fixed effects.

Migrants compared with residents of their birth township

Sample: born before 1953	College attendance or above (pp)	Professional/technical occupation (pp)	Household income rank (percentile)	Financial products held/invested in (pp)
Migrant	10.52^{**} (4.37)	8.56^{**} (3.11)	13.78^{**} (6.02)	14.38^{**} (5.51)
Observations	690	664	426	532

Migrant: current township differs from birth township. Survey weights; year, birth-year, and birth-township fixed effects. Birth-county clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Cash compensation and the migration threshold

Cash compensation relaxes the liquidity constraint on relocation. A lower migration threshold extends outflow beyond the top of the local hierarchy.

Migration by education

Outcome (pp of the origin sample)	D_i (per 10 pp)	Observations
Residence outside the birth township	50.196^{*} (26.966)	632
Outside birth township and below junior college	54.168^{**} (22.710)	632
Outside birth township and junior college or above	-3.972 (6.428)	632

TSCS 2012, 2017, 2022; born before 1953. All three outcomes use the same 632 respondents as the denominator. D_i: actual acquired dryland / 1950 township cultivated land, instrumented by Z_i. Ten historical controls; county, survey-year and birth-year FE. County CR1 standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; two-sided $t(G-1)$ tests.

As migration reaches less-educated groups, the migrant pool expands and its average schooling falls.

Outline

Historical Background and Literature Review

Data and Measurement

Short-run baseline population results

Estimating and controlling confounding factors from the land reform

Individual-Level Evidence on Migration

Long-run Outcomes

Conclusion

Appendix

Specification for long-run outcomes

$$y_{uit} = \alpha_{c(i)} + \lambda_t + \beta D_i^{(10)} + X_i'\gamma + \varepsilon_{uit}, \qquad D_i^{(10)} \equiv D_i/0.10.$$

D_i: actual redistributed dryland / 1950 township cultivated land, instrumented by Z_i (predicted dryland intensity). Effects are reported per 10 pp.

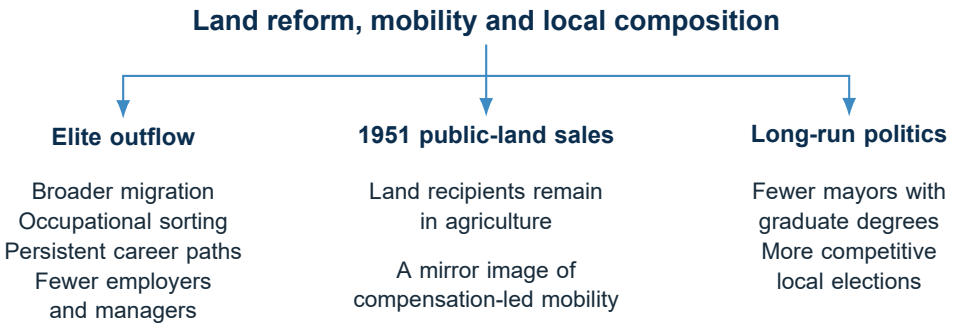
1950/1951 controls	Log 1951 population, log 1950 cultivated area, share of owners with at least 3 ha, and tenant-operated land share
1961 controls	Log crop output value and double-cropped paddy share
Colonial controls	Local Taiwanese headmen share, settlement diversity, Japanese colonizer land share, and colonial tenancy rate
Fixed effects	County and year. TSCS and all human-capital comparison columns also include birth-year fixed effects

Why the 1961 variables proxy conditions at reform

The literature finds little effect of land reform on grain output (Kim and Wang, 2025). Cultivated land had largely undergone modernization under Japanese rule, with little subsequent change after reform.

The 1961 variables control for persistent **natural endowments and irrigation infrastructure** not fully captured in the 1950 survey.

Conceptual roadmap: long-run consequences



Compensation-led mobility reshapes occupational paths, while public-land sales strengthen agricultural attachment.

Occupational sorting and career persistence

A. Initial industries of movers and stayers

First-job industry	Migrant − stayer (pp)	Observations
Manufacturing	17.102^{*} (8.143)	471
Agriculture	-21.544^{*} (10.556)	471

B. Initial industry and later work, at least ten years apart

Later-job industry (pp)	Same initial industry	Observations
Manufacturing	28.912^{***} (9.324)	375
Agriculture	26.558^{**} (9.243)	375

Born before 1953. A: TSCS 2012/2022; actual residence outside versus inside the birth township. B: TSCS 1992/1997/2002; initial job in 1953 or later, with at least ten years between job starts.

Both panels: survey weights; birth-township, survey-year and birth-year FE; birth-county CR1 standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Movers and stayers differ in their first jobs, and initial industry choices persist into later careers.

Agricultural lock-in and job mobility

Outcome (pp)	D_i (per 10 pp)	Observations	Years
A. Manpower Utilization Survey: linked annual records			
Agricultural-to-nonagricultural transition	-4.592 (3.905)	27,537	1979–2000
Enter nonagricultural self-employment or employer status	-0.946 (0.624)	74,408	1979–2000
Become an employer	-0.361 (0.432)	107,080	1979–2000
Change work township in the following year	-2.777^{**} (1.188)	101,760	1979–2000
B. Manpower Utilization Survey: annual survey			
Want to change or add a job	-1.599^{**} (0.569)	268,599	1978–2000

Coefficients: per 10 pp of actual dryland intensity; IV Z_i. County-clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Average social status

Taiwan Social Change Survey; born before 1953

Outcome	D_i (per 10 pp)	Observations	Years
Father attended junior college or above (pp)	-3.129^{*} (1.600)	536	2012, 2017, 2022
Self-rated social status (1–10)	-1.432^{***} (0.319)	537	2012, 2017, 2022
Main occupation: professional/technical (pp)	-13.222^{**} (4.879)	610	2012, 2017, 2022
Political-group participation (pp)	-4.379^{*} (2.262)	497	2012, 2022

Coefficients: per 10 pp of actual dryland intensity; IV Z_i. County-clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Human capital

Common workplace exposure; employed respondents born before 1953

Highest level attended (pp)	Manpower Utilization Survey	Taiwan Social Change Survey
Below primary school	-1.156 (3.590)	7.188 (6.182)
Primary school only	-0.582 (3.724)	-8.156 (11.719)
Junior high school only	0.978 (1.080)	-8.145 (7.865)
Senior high / vocational school only	-1.698 (2.096)	9.522^{**} (3.990)
Junior college or above	2.459 (2.033)	-0.409 (10.590)
Observations	158,082	1,041
Historical townships	210	150
County clusters	13	13
Exposure geography	Workplace	Workplace
Years	1978–2000	1992–2022

Five mutually exclusive education categories; each source uses its full eligible workplace sample. No cross-source township intersection. MUS birth year is approximated by survey year minus age.

2SLS: D_i instrumented by Z_i; effects per 10 pp. Ten historical controls; county, survey-year and birth-year FE in all columns. County CR1 SE; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Basic education and elite mobility

Broad educational attainment

The education coefficients do not show a significant decline in attainment among the local population.

Postwar expansion of compulsory education sustained broad access to basic schooling.

The top of the local hierarchy

- **Employers:** fewer local entrepreneurs
- **Managers:** a thinner managerial stratum
- **Mayors with graduate degrees:** fewer highly educated political elites

Migration broadens beyond the highly educated while employers and managers become less common in origin townships.

Employment structure and working hours

Outcome	Manpower Utilization Survey Workplace exposure	Internal Migration Survey Residence exposure
Private employees (pp)	-11.881^{**} (4.539)	-9.773^{*} (4.775)
Managerial occupations (pp)	-1.035^{**} (0.428)	-0.516 (0.448)
Finance, insurance, real estate, and business services (pp)	-0.670 (0.557)	-1.090^{*} (0.532)
Clerical occupations (pp)	-0.371 (1.690)	-4.522^{*} (2.346)
Manufacturing employment (pp)	-9.467^{*} (4.993)	-3.127 (5.179)
Secondary-sector employment (pp)	-11.164^{**} (5.075)	-7.149 (6.083)
Weekly working hours (hours)	2.311 (1.348)	2.713^{**} (1.157)
Observations (shares / hours)	268,978	127,221 / 114,817
Years	1978–2000	1980–1989, 1992

Per 10 pp of actual dryland intensity; IV Z_i. County-clustered SE in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.
Survey sampling weights; ages 25–64. IMS working hours: 1980–1989.

Public-land sales and employment

Outcome (pp)	Public-land share per 10 pp	Source	Observations
Agricultural employment, pooled censuses	2.242^{*} (1.051)	Five censuses	1,010
Agricultural occupations	1.880^{*} (0.968)	Four censuses	808
Agricultural industry	2.907^{*} (1.418)	Two censuses	404
Primary-sector employment	4.000^{***} (1.093)	MUS	268,978
Unpaid family workers	1.025^{**} (0.394)	MUS	268,978
Secondary-sector employment	-3.054^{**} (1.160)	MUS	268,978
Secondary-sector employment	-3.611^{**} (1.643)	IMS	127,221
Agricultural-to-nonagricultural transition	-1.090^{***} (0.235)	MUS linked	27,537
Change work township in the following year	-0.809^{***} (0.132)	MUS linked	101,760
Historical controls	Ten		
County and year FE	Yes		

MUS: Manpower Utilization Survey (workplace); IMS: Internal Migration Survey (residence). Census rows: OLS. Survey rows: private dryland intensity D_i instrumented by Z_i, with public-land share and predicted total redistribution as exogenous controls. Coefficients per 10 pp of leased public-land share. County CR1 SE in parentheses; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Township mayoral elections

Election sample

- Seven elections, 1998–2022
- 1,082 observations
- 210 historical townships in 13 counties

Outcomes

1. Mayor attended university or above
2. Mayor holds a graduate degree
3. Number of officially listed candidates
4. Winner’s share of valid candidate votes
5. Winner’s margin over the runner-up
6. Probability of a Kuomintang (KMT) victory

Township mayoral election outcomes

Actual redistributed dryland (2SLS); equal township-year weights

Outcome	D_i (per 10 pp)	Observations	Years
Mayor holds a graduate degree (pp)	-33.476^{***} (6.863)	310	2014–2022
Number of candidates	0.417^{**} (0.138)	1,082	1998–2022
Winner's vote share (pp)	-7.089^{**} (2.691)	1,082	1998–2022
Winner's margin over runner-up (pp)	-5.996^{**} (2.037)	985	1998–2022
Historical controls	Ten		
County and year FE	Yes		

Land-share coefficients: per 10 percentage points. Share outcomes: percentage points. Parentheses: historical-county CR1 clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$; two-sided $t(G-1)$ tests.

Actual dryland intensity D_i instrumented by predicted dryland intensity Z_i.

KMT support: elite outflow and land recipients

Land reform may affect KMT support through two channels: **elite outflow** and **gratitude toward the authorities or greater conservatism among land recipients**.

Channel	Explanatory variable	What it isolates
Elite outflow	$Q_i = A_i^D / A_i$; instrument: W_i	Cash orientation of acquired private land
Land recipients	$B_i = \dfrac{\text{1953 beneficiary households}}{\text{1951 agricultural households}}$	Exposure to receiving land

Dryland intensity D_i combines compensation type with the **extent of redistribution**. The dryland mix Q_i measures the composition of acquired land, instrumented by the 1950 township dryland share W_i.

The next table contrasts these channels on a common sample: the dryland-mix coefficient remains significant in the horse race, while beneficiary household share does not.

KMT victories: a horse race

$$KMT_{it} = \alpha_{c(i)} + \lambda_t + \beta Q_i^{(10)} + \theta B_i^{(10)} + X_i'\gamma + \varepsilon_{it}.$$

	(1) Dryland mix IV	(2) Beneficiary OLS	(3) Horse race IV
Actual redistributed dryland mix (10 pp)	2.748^{*} (1.580)	–	2.963^{*} (1.612)
Beneficiary household share (10 pp)	–	1.911 (2.016)	2.338 (2.031)
Historical controls	Ten	Ten	Ten
County and election-year FE	Yes	Yes	Yes
Observations	1,042	1,042	1,042
Historical townships	202	202	202
Township clusters	202	202	202
R^2	0.190	0.187	0.191

Outcome: KMT victory (pp). Same sample in all columns. Columns (1), (3): Q_i instrumented by W_i, the 1950 township dryland share; column (2): OLS. B_i: 1953 beneficiary households / 1951 agricultural households. Exposures per 10 pp. Township CR1 SE; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Horse-race p-values: actual dryland mix 0.068; beneficiary households 0.251.

Outline

Historical Background and Literature Review

Data and Measurement

Short-run baseline population results

Estimating and controlling confounding factors from the land reform

Individual-Level Evidence on Migration

Long-run Outcomes

Conclusion

Appendix

Conclusion

- **Mobility:** More liquid dryland compensation broadens migration beyond college-educated groups. Migrants have higher education, income rank, and financial-asset participation.
- **Local economy:** Movers and stayers differ in initial industries, and early industry choices persist. Origin townships have fewer employers and managers.
- **Local politics:** Instrumented actual redistributed dryland increases the number of candidates and lowers winners' vote shares and victory margins.
- **Public-land sales:** Greater public-land exposure reduces agricultural-to-nonagricultural transitions.

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Outline

Historical Background and Literature Review

Data and Measurement

Short-run baseline population results

Estimating and controlling confounding factors from the land reform

Individual-Level Evidence on Migration

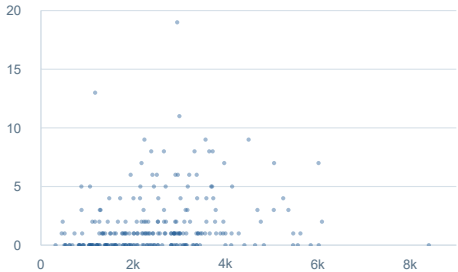
Long-run Outcomes

Conclusion

Appendix

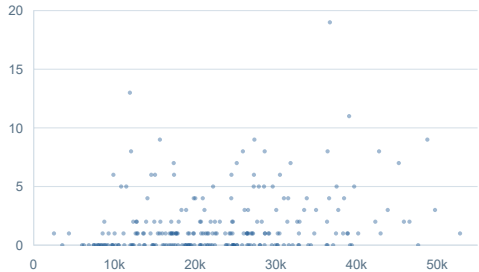
Large landowners and township size

1950 cultivated land (chia)



$r = 0.199$

1951 population (persons)



$r = 0.245$

Owner counts rise only modestly with township size. Population levels retain the direct interpretation in persons.

$N = 237$. Vertical axis: private-land owners with at least 30 hectares (about 30.9 chia). The 1950 survey reports hectares in bins; r denotes the Pearson correlation.

Step 1: observed inputs from the 1950 land survey

Private cultivated-land owners by holding-size bin

For township i, the survey reports n_{ib} owners in each of 12 ownership-area bins b:

$[0, 0.5)$, $[0.5, 1)$, $[1, 2)$, $[2, 3)$, $[3, 5)$, $[5, 7)$ ha
$[7, 10)$, $[10, 20)$, $[20, 30)$, $[30, 50)$, $[50, 100)$, $[100, \infty)$ ha

$$N_i = \sum_{b=1}^{12} n_{ib}$$

is the number of owners represented in the grouped distribution.

Other township-level fields

- total cultivated area C_i
- paddy area and dryland area Dry_i
- cultivated area by operating status, including tenant-operated paddy and dryland

Original source title: 《耕地所有權與經營狀況調查報告書》. The owner-size bins are measured in hectares; later calculations convert hectares and chia using $1 \text{ chia} = 0.969917$ hectares.

Step 2: fit a township owner-size distribution

Recommended distribution

$$F_i(a) = \Pr(A_i \le a) = \Phi\left(\frac{\ln a - \mu_i}{\sigma_i}\right)$$

County-bin shares $\pi_{c(i)b}$ supply limited shrinkage for small townships:

$$\widetilde{n}_{ib} = n_{ib} + \kappa \pi_{c(i)b}, \qquad \kappa = 25.$$

Parameters are estimated from the grouped likelihood:

$$(\widehat{\mu}_i, \widehat{\sigma}_i) = \arg\max_{\mu,\sigma} \sum_b \widetilde{n}_{ib} \ln[F_i(u_b) - F_i(l_b)]\,.$$

What the shrinkage does

- Adds 25 pseudo-owners distributed according to the 1950 county bin shares.
- Stabilizes the bin proportions of townships with few observed owners.
- Uses the 1950 land survey; N_i and fitted excess area determine each township estimate.

Model comparison

Five candidates are compared: lognormal, log-logistic, gamma, generalized gamma, and lognormal with a Pareto tail. The selected specification is lognormal with $\kappa = 25$.

The township likelihood uses the 1950 survey inputs. The 1953 county totals enter model comparison and validation.

Step 3: construct the dryland instrument

Expected land above a uniform threshold

Let $h = 3$ chia (2.909751 hectares). For one owner,

$$e_i = \mathbb{E}[(A_i - h)_+] = \int_h^\infty [1 - F_i(a)]\, da.$$

For a lognormal distribution,

$$e_i = \exp\left(\widehat{\mu}_i + \frac{\widehat{\sigma}_i^2}{2}\right) \Phi\left(\frac{\widehat{\mu}_i + \widehat{\sigma}_i^2 - \ln h}{\widehat{\sigma}_i}\right) - h\Phi\left(\frac{\widehat{\mu}_i - \ln h}{\widehat{\sigma}_i}\right).$$

Predicted township redistribution is

$$\widehat{R}_i = N_i e_i,$$

after converting the result back from hectares to chia.

Dryland component and normalization

$$q_i = \frac{Dry_i^{\text{tenant}}}{C_i^{\text{tenant}}}, \qquad \widehat{R}_i^D = \widehat{R}_i q_i,$$

$$Z_i = \frac{\widehat{R}_i^D}{C_i}.$$

Assume redistributed land has the same dryland share as tenant-operated cultivated land in 1950.

Z_i is the excluded instrument for actual dryland intensity D_i: predicted redistributed dryland divided by township cultivated land.

Validation against 1953 county totals

Construct	Predicted area	Actual area	Predicted / actual	Level corr.	Log corr.
Total redistributed land	122,313	139,003	0.880	0.852	0.923
Dryland: tenant land mix	30,204	21,385	1.412	0.800	0.886

